Brain Tasks DevOps Project

1. Project Overview

This project demonstrates a complete DevOps pipeline by deploying a React-based application on AWS using Docker, Amazon ECR, AWS CodeBuild, and Amazon EKS.

2. Architecture

GitHub → CodeBuild → Docker → Amazon ECR → Amazon EKS → LoadBalancer

3. Tools Used

GitHub, Docker, AWS CodeBuild, Amazon ECR, Amazon EKS, kubectl, eksctl, CloudWatch

4. Application Setup

Cloned application from GitHub and verified it runs on port 3000 using 'serve'.

5. Dockerization

Created Dockerfile using AWS public ECR nginx image to avoid Docker Hub rate limits.

6. ECR Setup

Created ECR repository and pushed Docker image using AWS CLI.

7. EKS Cluster Setup

Created EKS cluster using eksctl and configured kubectl to interact with cluster.

8. Kubernetes Deployment

Deployed application using deployment.yaml and service.yaml. Service type used: LoadBalancer.

9. CI/CD Pipeline

Configured AWS CodeBuild to automate Docker build and push to ECR.

10. Monitoring

Used AWS CloudWatch logs to monitor build and deployment.

11. Troubleshooting

Resolved issues like GitHub connection errors, IAM permission issues, Docker Hub rate limits, and ECR repository errors.

12. Final Output

Application URL: http://a70808154209647d79fbbc58b8d6935e-163265015.us-east-1.elb.amazonaws.com/

🚀 STEP 1: Launch EC2 (Your Main Work Machine)

👉 Create instance in Amazon EC2

Recommended:

- OS: Amazon Linux 2
- Instance: t3.medium
- Storage: 20GB

Security Group:

- Port 22 (SSH)
- Port 3000 (App testing)
- Port 80 (optional)

□ STEP 2: Install All Required Tools

```
sudo yum update -y

# Git
sudo yum install git -y

# NodeJS
curl -fsSL https://rpm.nodesource.com/setup_18.x | sudo bash -
sudo yum install -y nodejs

# Docker
sudo yum install docker -y
sudo systemctl start docker
sudo systemctl enable docker
sudo usermod -aG docker ec2-user
```

👉 Logout & login again after this step.

Verify: docker --version

AWS CLI

sudo yum install aws-cli -y

kubectl

curl -o kubectl https://s3.us-west-2.amazonaws.com/amazon-eks/1.27.0/2023-11-14/bin/linux/amd64/kubectl

chmod +x kubectl

sudo mv kubectl /usr/local/bin/

eksctl

curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz" | tar xz -C /tmp

sudo mv /tmp/eksctl /usr/local/bin

🚀 Step 3. Clone & Run Application (Port 3000)

git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
cd Brain-Tasks-App

```
[root@ip-172-31-79-176 ec2-user]# git clone https://github.com/Vennilavanguvi/Brain-Tasks-App.git
cd Brain-Tasks-App
Cloning into 'Brain-Tasks-App'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 11 (delta 0), reused 0 (delta 0), pack-reused 10 (from 1)
Receiving objects: 100% (11/11), 101.89 KiB | 20.38 MiB/s, done.
[root@ip-172-31-79-176 Brain-Tasks-App]#
```

Install serve

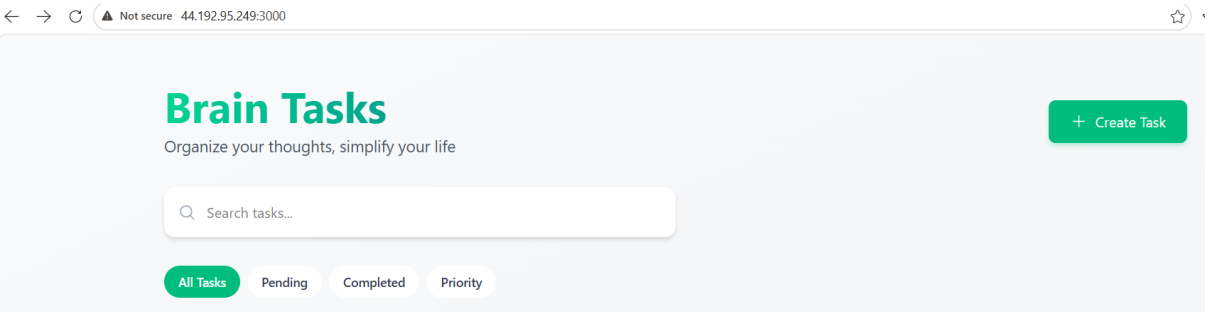
npm install -g serve

Run app

serve -s dist -l 3000

👉 Verify:

http://44.192.95.249:3000



🐳 STEP 4: Dockerize App

Create Dockerfile – Vi Dockerfile

FROM public.ecr.aws/nginx/nginx:alpine

COPY dist/ /usr/share/nginx/html/

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]

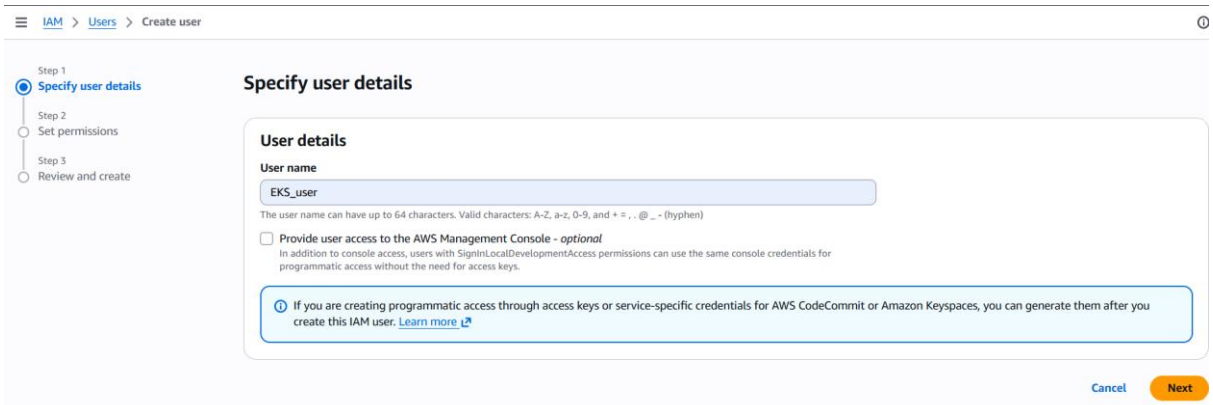
Build & Run

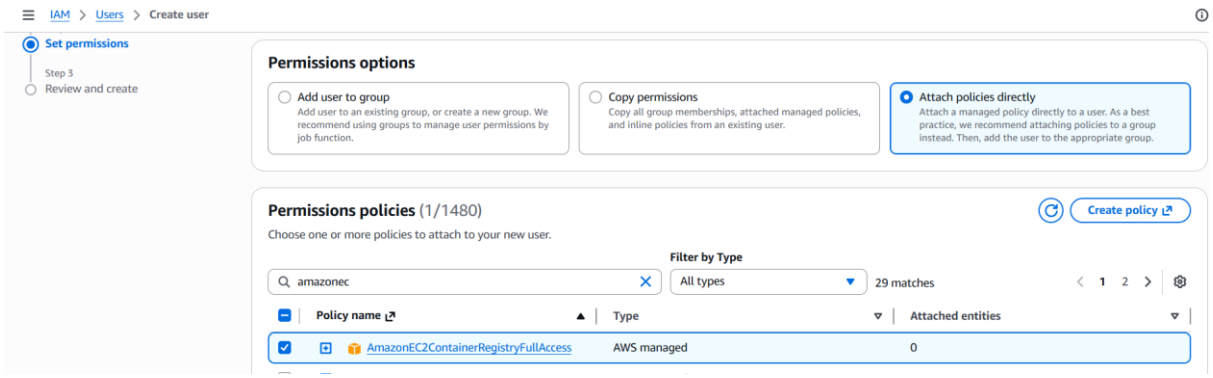
docker build -t brain-tasks-app .

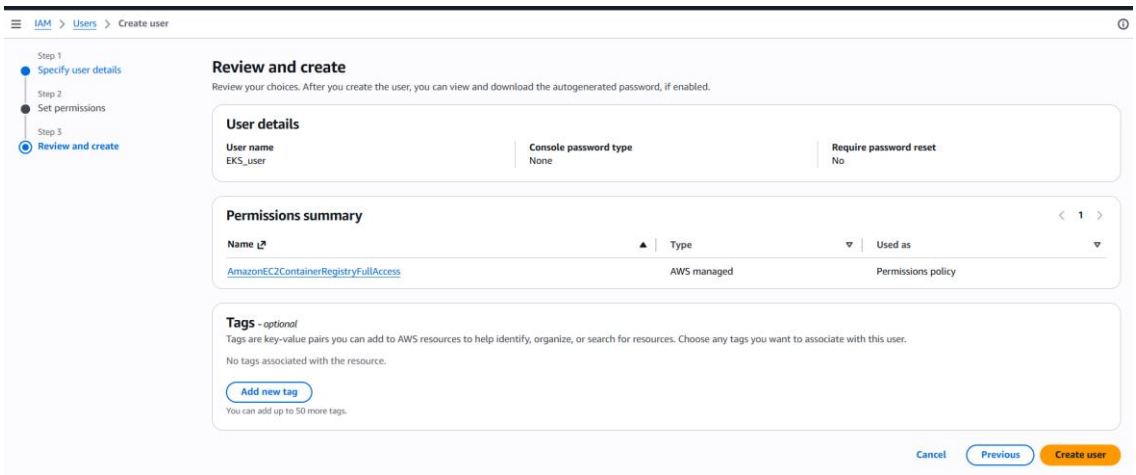
docker run -d -p 3000:80 brain-tasks-app

📦 STEP 5: Push to ECR

To Create a IAM user








```
[root@ip-172-31-20-145 Brain-Tasks-App]# aws configure
AWS Access Key ID [None]: AKIAZUJYQDXNBMLAZ36G
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
```

aws ecr create-repository --repository-name brain-tasks-app

aws ecr get-login-password --region us-east-1 | \

docker login --username AWS --password-stdin 662080331226.dkr.ecr.us-east-1.amazonaws.com

docker tag brain-tasks-app 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest

docker push 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest

```
[root@ip-172-31-20-145 ec2-user]# aws ecr get-login-password --region us-east-1 | \
docker login --username AWS --password-stdin 662080331226.dkr.ecr.us-east-1.amazonaws.com
docker tag brain-tasks-app 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app
docker push 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks app
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
"docker push" requires exactly 1 argument.
See 'docker push --help'.

Usage:  docker push [OPTIONS] NAME[:TAG]

Upload an image to a registry
[root@ip-172-31-20-145 ec2-user]# docker tag brain-tasks-app 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-ap
p:latest
[root@ip-172-31-20-145 ec2-user]# docker push 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
The push refers to repository [662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app]
4591a16375fd: Pushed
3d4a088932cf: Pushed
9f6dabb2aee2: Pushed
2fe86858894f: Pushed
0e7bffb1d445: Pushed
f365cb491667: Pushed
bdb037c8c9ed: Pushed
6e27cbd26acb: Pushed
989e799e6349: Pushed
latest: digest: sha256:8d1457a62ef706a1c0a8350499626b9de20153d5bb001a1e1ce5ade40faab707 size: 2199
[root@ip-172-31-20-145 ec2-user]#
```

STEP 6: Create EKS Cluster

eksctl create cluster \

--name brain-tasks-cluster-2\

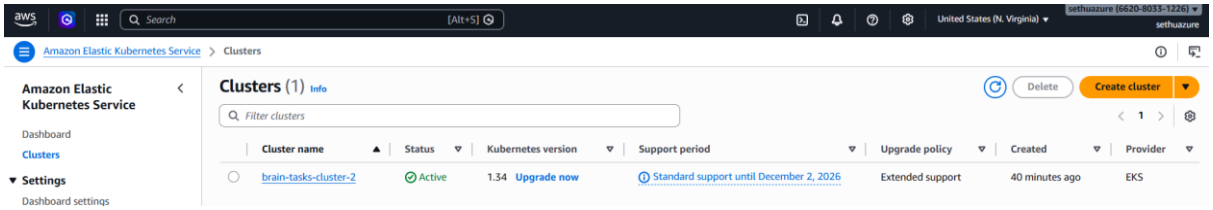
--region us-east-1 \

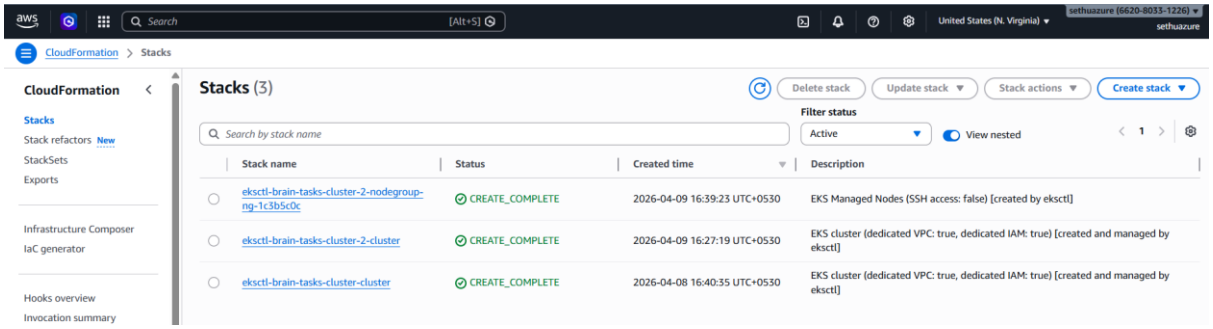
--nodegroup-name standard-workers \

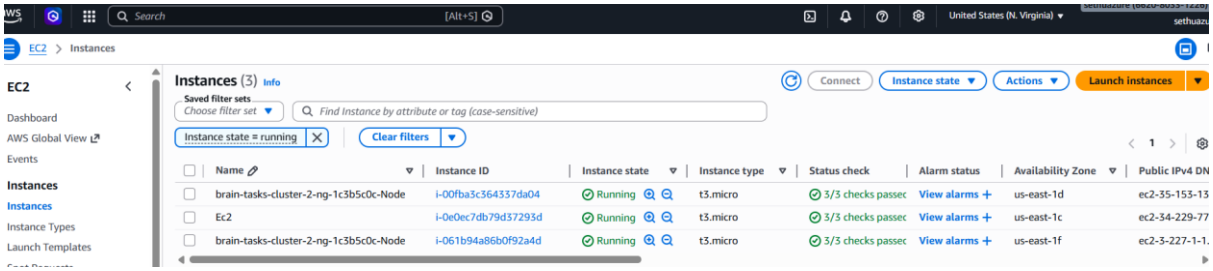
--node-type t3.small \

--nodes 2 \

--managed







Connect kubectl to cluster

aws eks update-kubeconfig \

--region us-east-1 \

--name brain-tasks-cluster-2

Verify nodes

kubectl get nodes

```
[root@ip-172-31-20-145 ec2-user]# aws eks update-kubeconfig \
--region us-east-1 \
--name brain-tasks-cluster-2
Added new context arn:aws:eks:us-east-1:662080331226:cluster/brain-tasks-cluster-2 to /root/.kube/config
[root@ip-172-31-20-145 ec2-user]# kubectl get nodes
NAME                              STATUS   ROLES    AGE   VERSION
ip-192-168-6-189.ec2.internal     Ready    <none>   28m   v1.34.4-eks-f69f56f
ip-192-168-62-184.ec2.internal    Ready    <none>   28m   v1.34.4-eks-f69f56f
[root@ip-172-31-20-145 ec2-user]#
```

📄 STEP 7: Deploy to Kubernetes

Vi deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: brain-tasks-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: brain-tasks
  template:
    metadata:
      labels:
        app: brain-tasks
    spec:
      containers:
      - name: brain-tasks
        image: 662080331226.dkr.ecr.us-east-1.amazonaws.com/brain-tasks-app:latest
        ports:
        - containerPort: 3000
```

Vi service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: brain-tasks-service
spec:
  type: LoadBalancer
  selector:
    app: brain-tasks
  ports:
    - port: 80
      targetPort: 80
```

kubectl apply -f deployment.yaml

kubectl apply -f service.yaml

```
[root@ip-172-31-20-145 ec2-user]# kubectl apply -f Deployment.yaml
deployment.apps/brain-tasks-deployment created
[root@ip-172-31-20-145 ec2-user]# kubectl apply -f Seployment.yaml
error: the path "Seployment.yaml" does not exist
[root@ip-172-31-20-145 ec2-user]# kubectl apply -f Service.yaml
service/brain-tasks-service created
[root@ip-172-31-20-145 ec2-user]# kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
brain-tasks-deployment-7b85874fb6-kwnlt   0/1     Pending   0          26s
brain-tasks-deployment-7b85874fb6-zv9t6   0/1     Pending   0          26s
[root@ip-172-31-20-145 ec2-user]#
```

👉 Get URL:

kubectl get svc

8. Version Control (GitHub)

git init

git remote add origin https://github.com/sethuraman98/brain-tasks-devops.git

```
git add .
git commit -m "Initial commit"
git push -u origin main
```

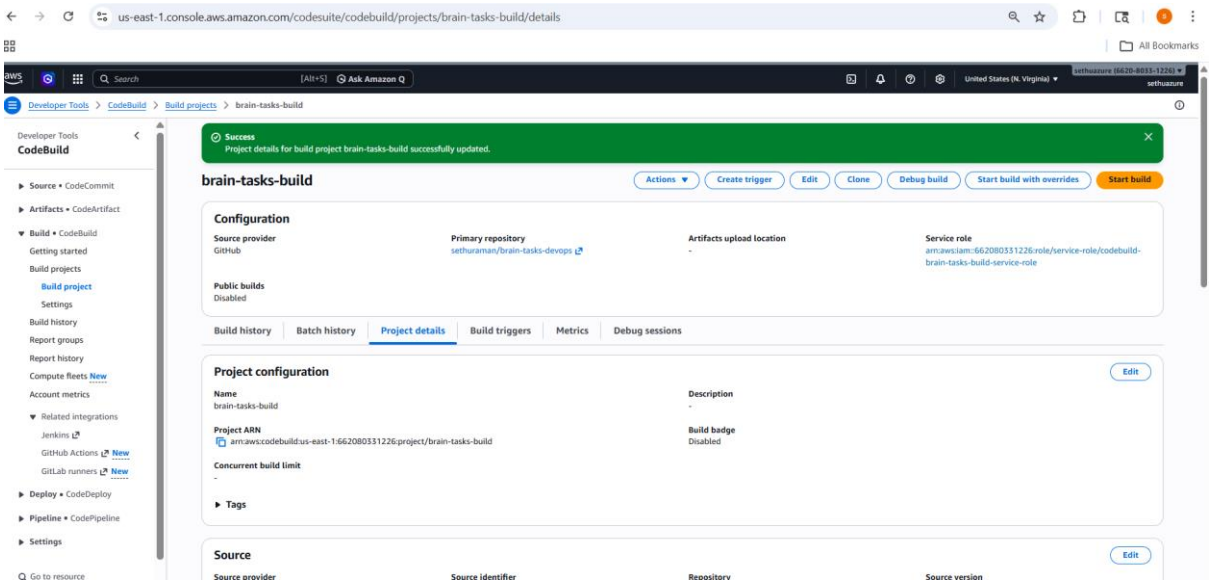
🔁 9. CodePipeline Setup

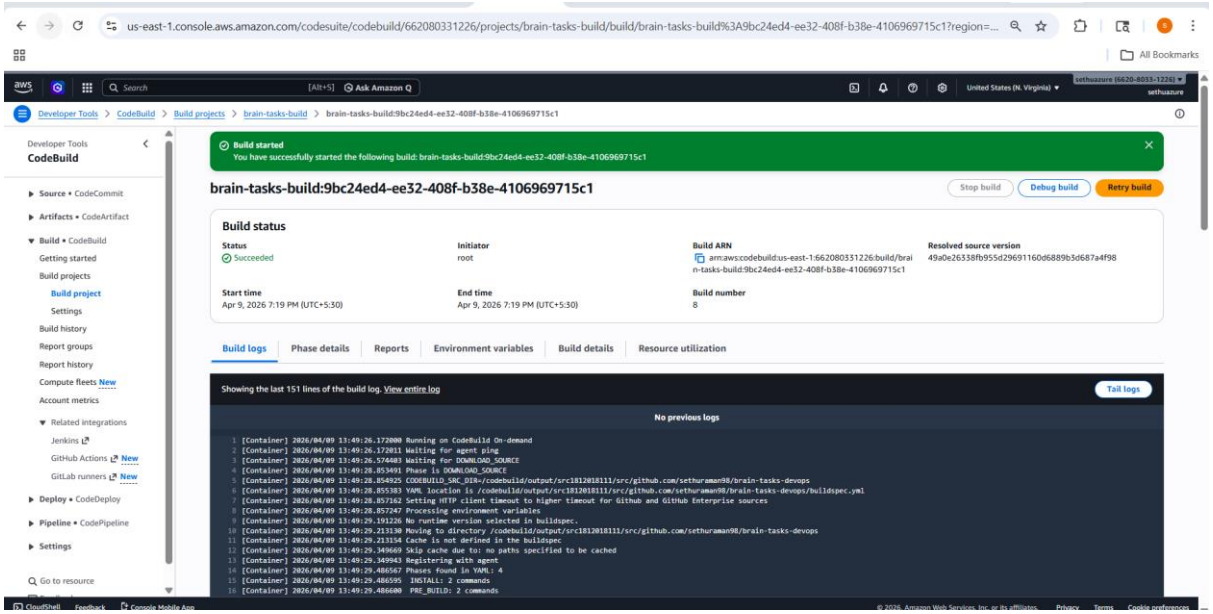
Pipeline Stages:

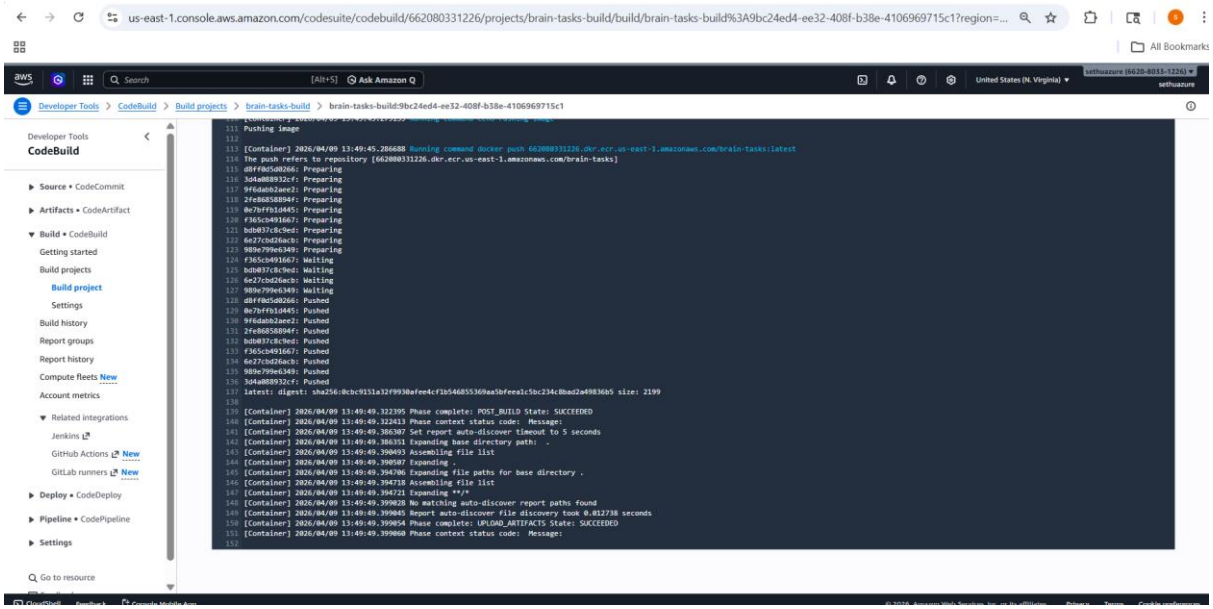
1. Source
   - GitHub repo
2. Build
   - AWS CodeBuild project
3. Deploy
   - Use CodeBuild or EKS deploy stage

👉 Ensure IAM roles allow:

- ECR access
- EKS access
- CloudWatch logs

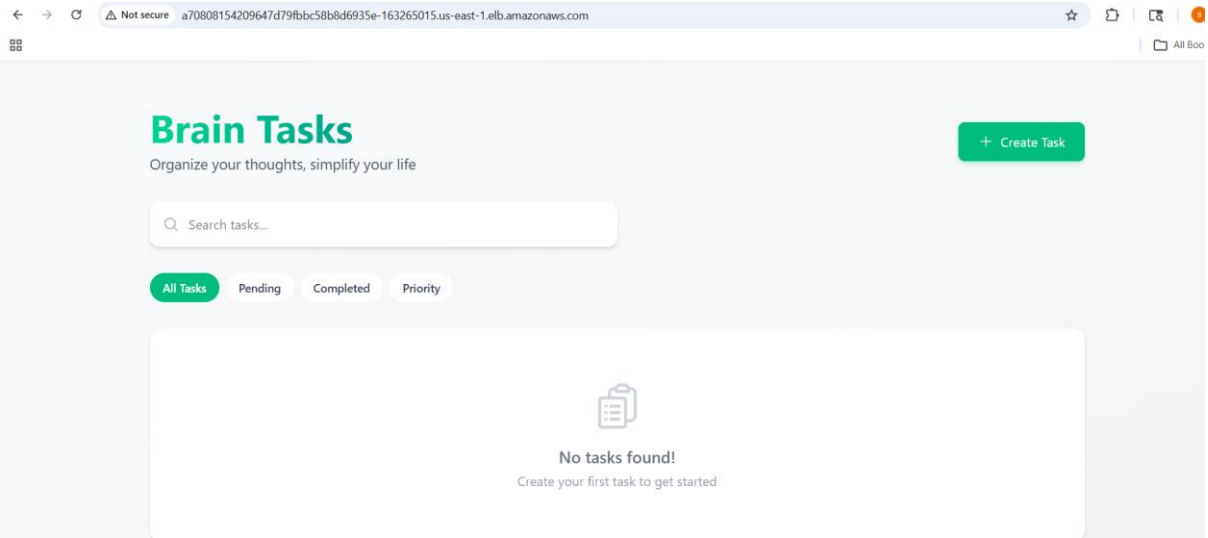




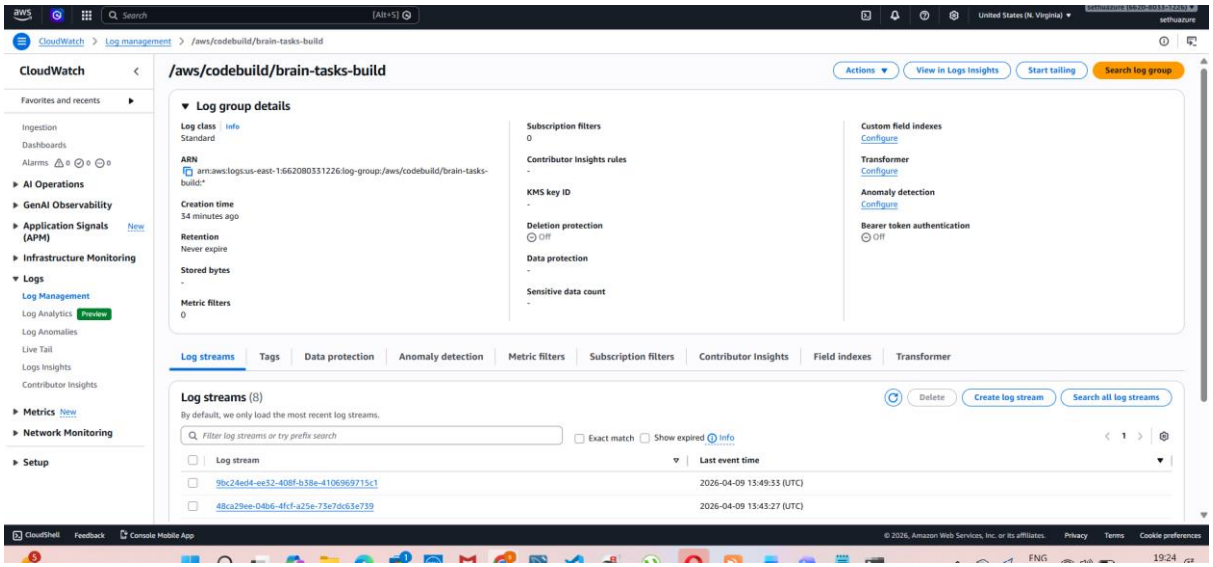


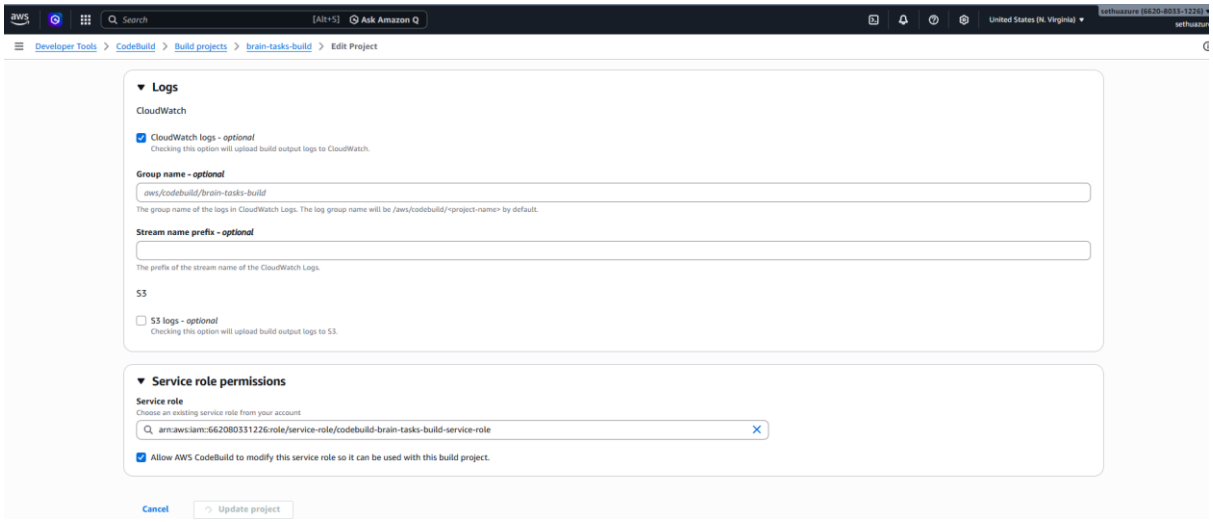
10. Application LoadBalancer URL:

http://a70808154209647d79fbbc58b8d6935e-163265015.us-east-1.elb.amazonaws.com/


Not secure
a70808154209647d79fbbc58b8d6935e-163265015.us-east-1.elb.amazonaws.com
Brain Tasks
Organize your thoughts, simplify your life
+ Create Task
Search tasks...
All Tasks
Pending
Completed
Priority
No tasks found!
Create your first task to get started

Cloudwatch


CloudWatch > Log management > /aws/codebuild/brain-tasks-build
CloudWatch
/aws/codebuild/brain-tasks-build
Actions
View in Logs Insights
Start tailing
Search log group
Log group details
Log class
Standard
ARN
arn:aws:logs:us-east-1:662080331226:log-group:/aws/codebuild/brain-tasks-build:*
Creation time
34 minutes ago
Retention
Never expire
Stored bytes
Metric filters
0
Subscription filters
0
Contributor Insights rules
KMS key ID
Deletion protection
Off
Data protection
Sensitive data count
Custom field indexes
Configure
Transformer
Configure
Anomaly detection
Configure
Bearer token authentication
Off
Log streams
Tags
Data protection
Anomaly detection
Metric filters
Subscription filters
Contributor Insights
Field indexes
Transformer
Log streams (8)
By default, we only load the most recent log streams.
Delete
Create log stream
Search all log streams
Log stream
Last event time
9bc24ed4-ee52-408f-b38e-4106969715c1
2026-04-09 13:49:33 (UTC)
48ca29ee-04b6-4fcf-a25e-73e7dc63e739
2026-04-09 13:43:27 (UTC)



“I implemented a full DevOps CI/CD pipeline using AWS services including Docker, ECR, EKS, and CodeBuild to deploy a React application, which is accessible via a LoadBalancer URL.”